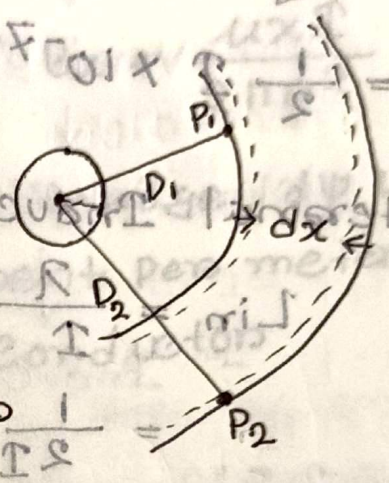
Art - 2.3: Flux Linkage Due to Flux between two points external to Conductor

Two points P_1 and P_2 at distance D_1 and D_2 from a conductor I which carries a current I. Magnetic field external to the conductor is concentric circles around the conductor.



Magnetic flux at the tabular element,

$$d\phi = \frac{\mu I}{2\pi x} \cdot dx$$

Total flux linkage between P_1 and P_2

$$\lambda_{12} = \int_{D_1}^{D_2} \frac{\mu I}{2\pi x} dx$$

$$= \frac{\mu I}{2\pi} \int_{D_1}^{D_1} \frac{dx}{x}$$

$$= \frac{\mu I}{2\pi} \ln(x) \Big|_{D_1}^{D_2}$$

$$= \frac{\mu I}{2\pi} \ln \frac{D_2}{D_1} = \frac{\mu_0 \mu_r I}{2\pi} \ln \frac{D_2}{D_1}$$

$$\lambda_{12} = \frac{4\pi \times 10^{-7}}{2\pi} I \ln \frac{D_1}{D_2}$$

$$= 2 \times 10^{-7} I \ln \frac{D_1}{D_2}$$

∴ Inductance of the conductor included between P_1 and P_2.

$$\left[L_{12} = 2 \times 10^{-7} \ln \frac{D_2}{D_1} \quad H/m \right]$$

$$= 0.461 \log \frac{D_2}{D_1} \quad mH/km$$

Flux linkage due to flux upto a external point.

Let, external point at distance D from the center of the conductor.

$D_2 = D, \quad D_1 = r$

$$L_{12} = 2 \times 10^{-7} \ln \frac{D}{r}$$

Total flux linkage (due to the flux upto an external point),

$$\lambda = \lambda_{in} + \lambda_{ext}$$

$$= \frac{1}{2} I \times 10^{-7} + 2 \times 10^{-7} I \ln \frac{D}{r}$$

$$= 2 \times 10^{-7} I \ln \left[1/4 + \ln \left(\frac{D}{r} \right) \right]$$

$$= 2 \times 10^{-7} I \ln \left(\frac{D e^{1/4}}{r} \right)$$

$$\lambda = 2 \times 10^{-7} \; I \ln\left(\frac{D}{e^{-1/4} r}\right)$$

Here, $r' = 0.7788\, r = e^{-1/4} r$

$$\lambda = 2 \times 10^{-7} I \ln\left(\frac{D}{r'}\right) \text{ wb. T/m}$$

[we must use r'.]

$$\boxed{L = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right) \text{ H/m}}$$

Ant-2.4 Inductance of a Single phase two wire line

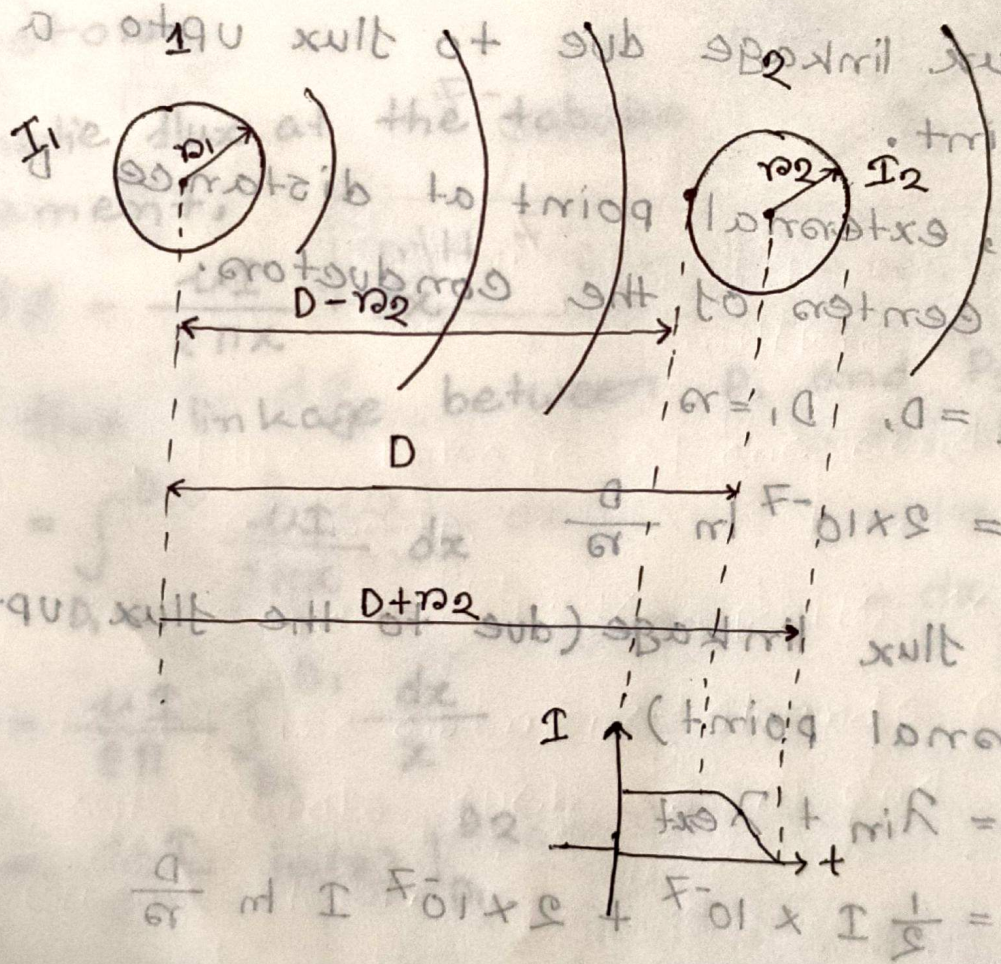


Fig: Single phase two wire line and the magnetic field due to current in conductor 1 only.

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Consider Simple two wire line composed to solid round Conductor carrying currents I_1 and I_2. In a Single phase line, $I_1 + I_2 = 0$

$\Rightarrow I_1 = -I_2$

Now, flux linkages of the circuit caused by current in Conductor 1 only, we need three assumptions.

① External flux from r_1 to $(D - r_2)$ links all current I_1 in Conductor '1'

② External flux from $D - r_2$ to $D + r_2$ links a current where magnitude progressively reduces from I_1 to zero along this distance because of negative current flowing in Conductor 2.

③ Flux beyond $(D_1 + r_2)$ links a net current to zero.

④ So, flux linkage due to conductor 1 only $\lambda_1 = 2 \times 10^{-7} I_1 \ln \frac{D}{r_1}$

Inductance of the Conductor due to current in conductor 1

$$L_1 = 2 \times 10^{-7} \ln \frac{D}{r_1}$$

Similarly, for conductor 2,

$$L_2 = 2\times10^{-7} \ln \frac{D}{r_2'}$$

Total Inductance $L = L_1 + L_2$

$$= 2\times10^{-7} \ln \frac{D^2}{r_1' r_2'}$$

$$= 2\times10^{-7} \ln \left(\frac{D}{\sqrt{r_1' r_2'}}\right)^2$$

$$L = 4\times10^{-7} \ln \left(\frac{D}{\sqrt{r_1' r_2'}}\right)$$

If $r_1' = r_2' = r'$

$$L = 4\times10^{-7} \ln \frac{D}{r'} \quad H/m$$

$$= 0.921 \log \frac{D}{r'} \quad mH/km$$

Art-2.6: Flux linkage of one conductor in a group.

Composite Conductor: Composite conductor consists of two or more strands of different metals such as Al and steel or Cu or steel.

Consider, a composite conductor, which has two or more strands of diff. metals connected in parallel. All the strands are identical and equal current sharing. such that the currents are zero.

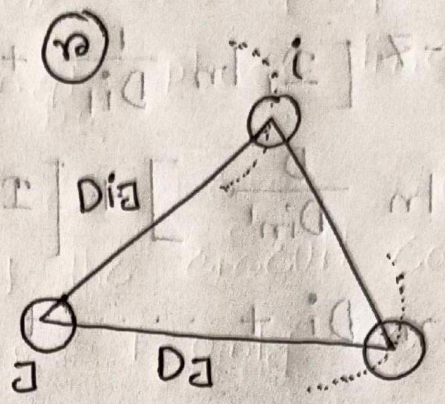


Fig: Arbitrary group of n parallel round conductors carrying currents.

From book eqn. 2.21

$$\lambda = 2\times10^{-7} I \ln \frac{D}{r'}$$

$$\lambda_{ij} = 2\times10^{-7} I_i \ln \frac{D_i}{r_i'} \quad - \text{own}$$

$$\lambda_{12} = 2\times10^{-7} I \ln \frac{D_2}{D_1} \quad - \text{From book eqn. 2.17}$$

Flux linkages of ith conductor due to current in conductor j

$$\lambda_{ij} = 2\times10^{-7} I_j \ln \left(\frac{D_{ij}}{D_{ii}}\right) \text{ wb. t/m}$$

Total flux linkage of conductor i

$$\lambda_i = \lambda_{i1} + \lambda_{i2} + \lambda_{i3} + \cdots + \lambda_{ii} + \cdots + \lambda_{in}$$

$$= 2\times10^{-7}\left[I_1 \ln \frac{D_1}{D_{i1}} + I_2 \ln \frac{D_2}{D_{i2}} + \cdots + \right.$$

$$I_i \ln \frac{D_i}{r_i'} + \cdots + I_n \ln \frac{D_n}{D_{in}} \Big]$$

$$= 2\times10^{-7} \Big[I_1 \ln \frac{1}{D_{i1}} + I_2 \ln \frac{1}{D_{i2}} + \cdots + I_2 \ln \frac{1}{r_i'}$$

$$+ I_n \ln \frac{D}{D_{in'}} \Big] + \Big[I_1 \ln D_1 + I_2 \ln D_2 + \cdots$$

$$+ I_i \ln D_i + \cdots + I_n \ln D_n \Big]$$

The sum of current is zero

$$I_1 + I_2 + \cdots + I_i + \cdots + I_n = 0$$

$$I_n = -(I_1 + I_2 + I_i + \cdots + I_{n-1})$$

$$\lambda_i = 2\times10^{-7} \Big[I_1 \ln \frac{1}{D_{i1}} + I_2 \ln \frac{1}{D_{i2}} + \cdots +$$

$$I_n \ln \frac{D}{D_{in}} \Big] + \Big[I_1 \ln D_1 + I_2 \ln D_2$$

$$+ \cdots + I_i \ln (D_i) + \cdots + (-I_1 - I_2 - \cdots - I_{n-1})$$

$$\ln D_n \Big]$$

$$\lambda_i = 2\times10^{-7} \Big[I_1 \ln (1/D_{i1}) + I_2 \ln \frac{1}{D_{i2}} + \cdots$$

$$+ I_i \ln \frac{1}{r_i} + \cdots + I_n \frac{1}{D_{in}} \Big] + \Big[I_1 \ln \frac{D_1}{D_n}$$

$$+ I_2 \ln \frac{D_2}{D_n} + I_i \ln \Big(\frac{D_i}{D_n} \Big) + \cdots + I_{n-1}$$

$$\ln \frac{D_{n-1}}{D_n} \Big]$$

* $I_1 \ln D_1 - I_2 \ln D_2 = I_1 \ln \frac{D_1}{D_n}$

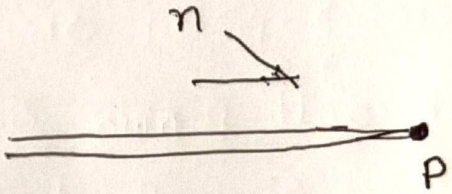
* $I_2 \ln D_2 - I_2 \ln (D_n) = I_2 \ln \frac{D_2}{D_n}$

If point P is at a infinite distance

$$\frac{D_1}{D_n} = 1 ; \quad \frac{D_2}{D_n} = 1$$

$$\ln \frac{D_1}{D_n} = \ln(1) = 0$$

$$\therefore \lambda_i = 2 \times 10^{-7} \left[I_1 \ln \frac{1}{D_{i1}} + I_2 \ln \left(\frac{1}{D_{i2}}\right) + \cdots + I_i \ln \left(\frac{1}{r_{ii}}\right) + \cdots + I_n \ln \left(\frac{1}{D_{in}}\right) \right] \quad wb\text{-}T/m$$



* $I_1 \ln D_1 - I_2 \ln D_2 = I_1 \ln \frac{D_1}{D_n}$

* $I_2 \ln D_2 - I_2 \ln (D_n) = I_2 \ln \frac{D_2}{D_n}$

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